toys10_07-tilt-maze

Tilt Maze Labyrinth: manufacturing manual

> **Status: design only; nothing has been cut or assembled.**

1. What is made

1. Turn two side knobs to tilt the maze; roll the marble from start to goal without falling in the trap holes.
2. Three interchangeable maze plates (9×9, 13×13, 17×17); race a friend by timing.

2. Files and materials

- Maze plates: `toys10/out/maze_easy.svg` (9x9), `maze_medium.svg` (13x13), `maze_hard.svg` (17x17): perfect mazes, trap holes on dead ends only, 19 mm corridors for a 12 mm marble; checks `toys10/docs/results_maze.md`
- Gimbal (base, two uprights, finger-jointed middle frame, tray for the plates, axle bosses, knobs): `toys10/out/tilt_maze_gimbal_sheet1.svg`/`.dxf` and `_sheet2` (3 mm MDF, kerf-compensated, each sheet fits a 900 x 600 mm bed), generated by `toys10/sims/gen_gimbal.py`; geometry checks, axle lengths and assembly order in `toys10/out/tilt_maze_gimbal_checks.md` (all PASS). First-pass: cut one set in scrap MDF and test the tilt before a batch.
- Material: **3 mm MDF or acrylic (plates with glued 3 mm walls)**; layout/sheet: per SVG.

Cost per set (about 2,000 units, estimate):

Item	Rs
Laser-cut frame + 3 plates	110
Wall strips	25
Knobs + dowels + bearings	40
Marbles ×3	10
Box	45
Assembly	30

3. Making it

Follow `manuals/STANDARD_PROCEDURES.md` section H (laser cutting) for kerf tests and finishing, then:

1. Cut the three plates; glue the 3 mm wall strips to the plates along the maze lines (PVA, clamp, wipe squeeze-out; corridors must stay 19 mm wide: check with a 12 mm marble).
2. **Design and cut the gimbal frame** (not available): outer frame on a stand, inner frame carrying the interchangeable plate, two side knobs on 6 mm dowels; make a cardboard mock-up first and record tilt range and friction.
3. Fit marbles (12 mm or larger: choking hazard, age 7+), check that no marble can jam in a trap hole.

4. Sand, seal, assemble knobs and bearings.

4. Inspection (sample per batch)

- Every plate: marble can reach the goal; traps only on dead ends (verified by BFS in the design).
- Knobs tilt the board smoothly; marble rolls without sticking.
- Marble choking-hazard label present (age 7+).

Record on `manuals/templates/eol_test_sheet.md`.

5. Packaging and compliance

Toys QCO / IS 9873 applicability and testing; age grade and small-parts warnings; Legal Metrology labelling; paint/ink heavy-metal report; no licensed names or logos.

Add the age grade, small-parts warning, Legal Metrology marks and a printed instruction leaflet.

6. Risks and limits

- Tilt feel and friction not tested.
- Marble choking hazard: marble 12 mm or larger, 7+ label.

Marble is a choking hazard: use age 7+ labelling and a marble ≥ 12 mm.

`docs/results_maze.md`: all three mazes are perfect (one solution, BFS-checked); trap holes sit on dead ends only, so no hole blocks the solution; corridor 19 mm for a 12 mm marble. Not tested: tilt feel and friction.